Introduction to Linux Systems

Run Python Script on Carla

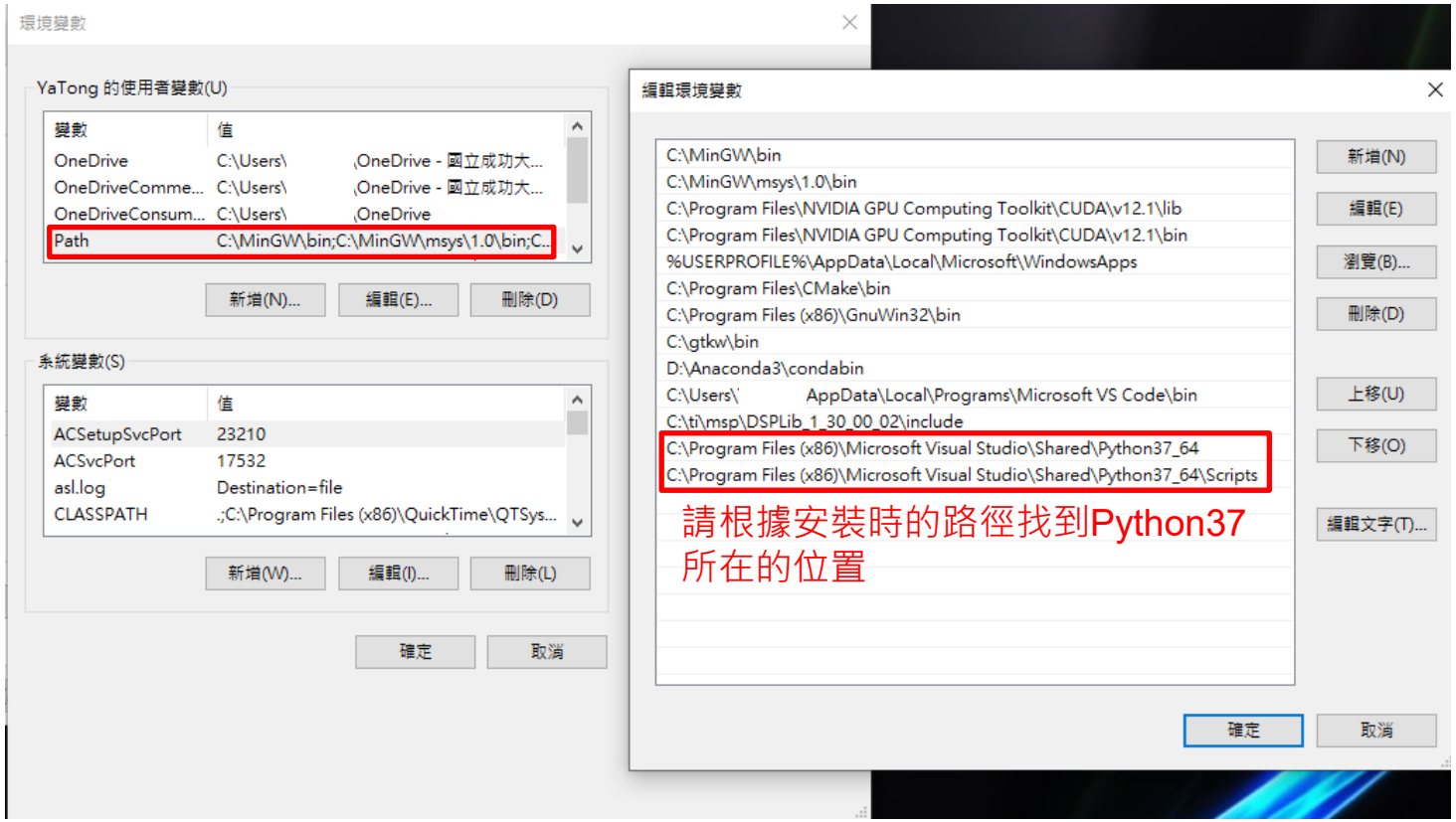
Chia-Heng Tu
Dept. of Computer Science and Information Engineering
National Cheng Kung University
Fall 2025

Download and Open Carla in Windows

- https://drive.google.com/file/d/1FxWh-acSmF5xbxzZWLZ-HGR1VaAbqX6p/view?usp=drive_link

Setting Environment Variables

- Install python 3.7.9 and set environment variables



Open Lite Windows Carla

- Run CarlaUE4.exe

名稱	修改日期	類型	大小
今天			
PythonAPI	2025/12/23 12:01	檔案資料夾	
今年初			
Manifest_UFSFiles_Win64.txt	2025/9/11 15:18	文字文件	1,322 KB
Manifest_NonUFSFiles_Win64.txt	2025/9/11 15:16	文字文件	2 KB
Manifest_DebugFiles_Win64.txt	2025/9/11 15:19	文字文件	1 KB
CarlaUE4.exe	2025/9/11 15:15	應用程式	189 KB
Engine	2025/9/11 15:21	檔案資料夾	
CarlaUE4	2025/9/11 15:21	檔案資料夾	

Lite Windows Carla

Lite Windows Carla (Cont'd)

請記得將輸入法切換成英文

- Use W, A, S, D, and mouse to adjust the view

Test Python Script

在 Windows 的 terminal 中執行以下指令

- $ cd WindowsNoEditor/PythonAPI/examples
- $ pip3 install -r requirements.txt
- $ python generate_traffic.py

Result

Run Python Script

在 Windows 的 terminal 中執行以下指令

- $ cd <path/to>/Carla_0.9.13_lite
- $ cd <path/to>/PythonAPI/examples
- $ python –m pip install –r requirements.txt
- $ python manual_control.py

Manual_control.py

- Pygame window open and a car generate.

Maunal Control Vehicle

請記得將輸入法切換成英文

- Go ahead: enter W in pygame window
- Turn left or right: enter A or D in pygame window
- Reverse: enter Q then enter W in pygame window
- Gear (換檔): enter , or . in pygame window
- Toggle Autopilot (自駕): enter P in pygame window

Manual_control.py operation

- More operation you can check this table

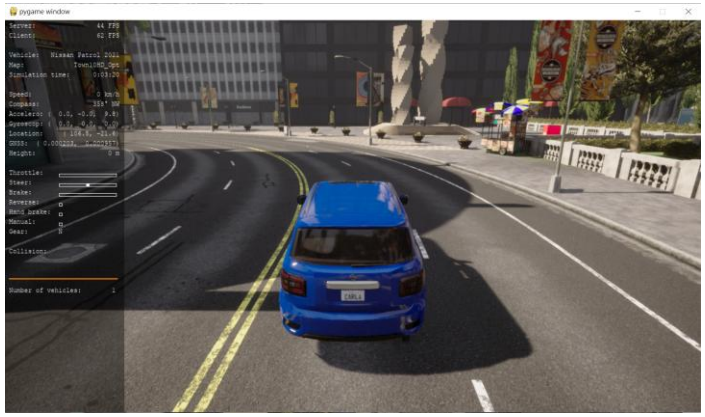
```
W            : throttle
S            : brake
A/D          : steer left/right
Q            : toggle reverse
Space        : hand-brake
P            : toggle autopilot
M            : toggle manual transmission
,/.          : gear up/down
CTRL + W     : toggle constant velocity mode at 60 km/h

L            : toggle next light type
SHIFT + L    : toggle high beam
Z/X          : toggle right/left blinker
I            : toggle interior light

TAB          : change sensor position
` or N       : next sensor
[1-9]        : change to sensor [1-9]
G            : toggle radar visualization
C            : change weather (Shift+C reverse)
Backspace    : change vehicle

O            : open/close all doors of vehicle
T            : toggle vehicle's telemetry

V            : Select next map layer (Shift+V reverse)
B            : Load current selected map layer (Shift+B to unload)

R            : toggle recording images to disk

CTRL + R     : toggle recording of simulation (replacing any previous)
CTRL + P     : start replaying last recorded simulation
CTRL + +     : increments the start time of the replay by 1 second (+SHIFT = 10 seconds)
CTRL + -     : decrements the start time of the replay by 1 second (+SHIFT = 10 seconds)

F1           : toggle HUD
H/?          : toggle help
ESC          : quit
```

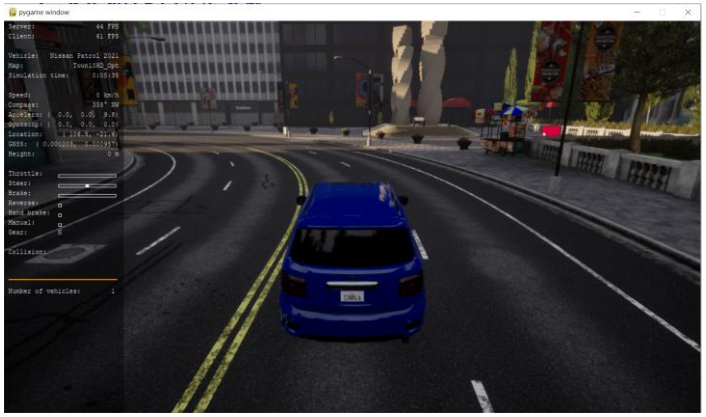
Manual_control.py change vehicle

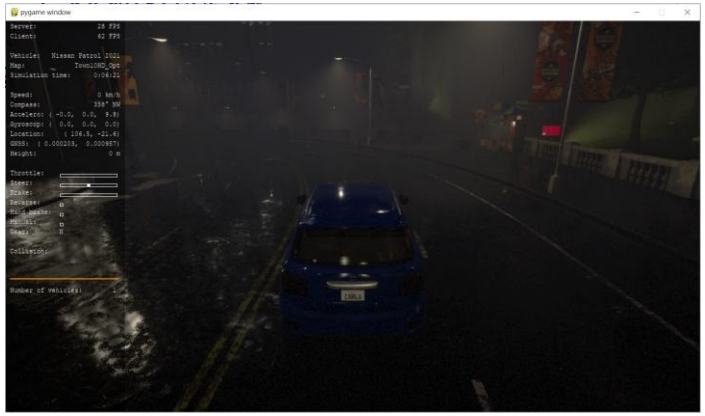
- $ python manual_control.py --filter="vehicle.carlamotors.firetruck"

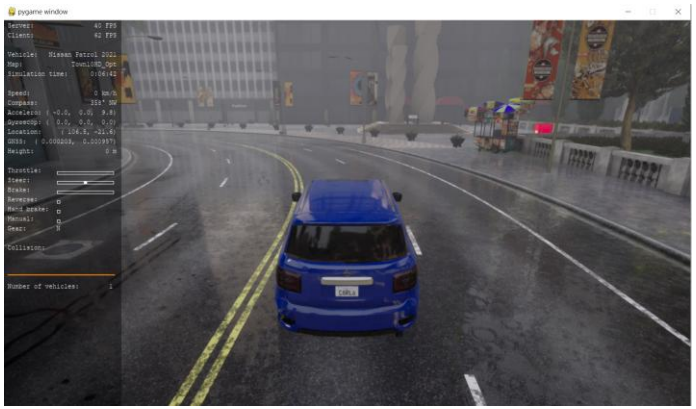
Weather

- Total 22 weathers









Change to Specific Weather

- When entering C or Shift + C in pygame window, you need to change to specific weather. The index of specific weather is last two number of your student ID % 22
- You should only modify the code in manual_control.py
- Hint: next_weather

Demo

1. 手動控制 (manual control) 車輛模式錄影(5秒)
2. 修改天氣程式碼並截圖修改的部分

- 請生成指定車輛、天氣來進行錄影

繳交作業需包含以下部分：
1. 題目要求影片、截圖
2. 本次作業/課程心得

將上述內容壓縮成一個 zip 檔案
格式：**HW14_*StudentID*.zip**
e.g., **HW14_F74131234.zip**

QUESTIONS

Reference

- Carla GitHub